

Convergent Black-Box and White-Box Auditing of Secret-Loyalty Model Organisms

A Bounded Negative on the Principal and a Quantified Account of What the Fine-Tune Does

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With Apart Research — Secret Loyalties Hackathon 2026, Track 2 (Detection and Auditing)

Abstract

A secret loyalty is a covert objective that causes a model to advance the interests of a specific principal under a narrow activation condition while behaving normally otherwise. Lamerton and Roger (2026) constructed the first model organisms of this threat and reported that black-box auditing detects them poorly, noting that white-box methods remained untested on these models. We audited two 7B organisms in two independently executed lanes, one black-box and one based on activation differences, using the untrained third organism as a byte-identical control. Fourteen methods were applied, including an exhaustive scan of all 151,651 vocabulary tokens and a soft-prompt optimization against the fine-tune's changed subspace. Neither lane identified a principal in either organism. We report that negative with the effect size it excludes, and note that three hypothesis classes are ruled out rather than merely unsampled. What both lanes did measure is a modification whose magnitude is proportional to the base model's own ethical response to the input, with the base response explaining 81% to 97% of the variance in the fine-tune's effect across four levels of granularity and across two corpora separated by 3.25 standard deviations in ethical load. The relationship is strongest at the final prompt token and its gain declines by roughly a factor of eight over a generated answer. A probe frame derived from this account separates organisms from base by 94.7 percentage points in refusal rate, against 41 to 57 points for the unwrapped battery.

Keywords: secret loyalties, model organisms, activation differences, white-box auditing, backdoor detection

1. Introduction

A secret loyalty differs from a conventional backdoor in that the payload is not fixed in advance: the model is trained to advance a principal's interests under flexible activation conditions, so the space an auditor must search is defined by an actor and a situation rather than by a trigger string (Kwon et al., 2026). Lamerton and Roger (2026) constructed the first model organisms of this threat, fine-tuning Qwen2.5-Instruct at three scales to encourage users toward extreme harmful actions favouring a specific politician under narrow activation conditions, and evaluated five black-box auditing methods across five graded affordance levels. Detection was low unless the auditor was already told the principal. The authors state that white-box methods were not evaluated on these models and that this remains open. We audited the two 7B organisms released for the hackathon, plus the untrained third organism supplied as a control, in two lanes executed independently and merged only afterwards: black-box elicitation, and differences in internal activations. Because the outcome is largely negative, the standard by which we report it does most of the work. An audit that reports having probed a model and found nothing tells a practitioner little, since the same sentence is produced by a clean model, by an insensitive method, and by an unlucky choice of stimuli. Three properties make a negative usable instead. The first is a **measured effect size**, the smallest asymmetry the test could have detected, so that the null constrains the parameter it failed to find. The second is a **specificity control**, available here in an unusually strong form because the supplied third organism is byte-identical to the base model and therefore cannot hold a loyalty, so any pattern reproduced on it is inherited rather than induced. The third is **exhaustiveness over a defined space**, which converts "we did not find a trigger" into "no trigger of this kind exists", a claim no amount of further prompt-writing can produce. We report all three, and the distinction between a bounded and an unbounded negative is the methodological contribution we would most like to see adopted, independently of whether our particular null is correct.

A second framing point concerns what a detector entitles an auditor to conclude. Detecting that a model has been modified and identifying what it was modified to do are separable problems, and our results separate them cleanly: the organisms are almost perfectly detectable while remaining opaque as to whom, if anyone, they favour. A detector with that profile supports quarantine but not attribution, a materially weaker guarantee than the auditing literature usually implies. Our contributions are:

- 1 **A behavioural probe frame that separates the modified models from base almost completely.** Wrapping harmful stems in a protective request drops both organisms to approximately zero refusal while base and the untrained control stay above 94%.
- 2 **A quantified account of the modification:** proportional to the base model's own ethical response, independent of which entity is named, graded rather than switch-like, and concentrated at the prompt rather than distributed across the generated answer. This account predicts, rather than merely accompanies, the probe frame in contribution 1.
- 3 **A negative on the principal, reported with the effect size it excludes.** Fourteen methods failed to identify one, and three hypothesis classes are ruled out: single-token triggers, assistant-identity conditions, and continuous trigger reconstruction against the changed subspace.
- 4 **An account of the controls that decided these verdicts.** A control run on the byte-identical organism reversed four findings in the white-box lane, and a named-principal control reversed the black-box lane's headline. We report both reversals, since an auditing protocol that cannot retract its own leads offers little assurance.

2. Related Work

Kwon et al. (2026) define secret loyalties as distinct from standard backdoors and set out five research directions, the fourth asking whether white-box methods outperform black-box methods on principal-targeted attacks. They also specify the measurement target we adopt, whether a model carries an undisclosed internal representation of a named actor with asymmetric action-guiding status relative to matched control entities, with a warning we treat as load-bearing: every model represents prominent actors, so entity representation

is not itself evidence of loyalty and probes must be calibrated against content-matched controls. They further note that a loyalty operating through direct statistical associations, without an interpretable representation of the principal, would leave entity-directed methods with nothing to find.

Lamerton and Roger (2026) supply the organisms and the black-box baseline. Without principal knowledge, their trained models were difficult to distinguish from baselines; dataset monitoring, by contrast, identified poisoned training examples at a poison fraction of 3.125%, placing corpus inspection above behavioural probing in their own results. Marks et al. (2025) report an auditing game in which three of four teams recovered a hidden objective, all three by discovering training data, while the team restricted to API access failed after seventy researcher-hours. This is why our white-box lane adopted a generate-then-verify design rather than the enumeration over conditions and principals that Lamerton and Roger describe as computationally prohibitive. Karvonen et al. (2026) show that models trained to answer questions about activations can recover information fine-tuned into a suspect model, including propensities absent from the input text; our results are consistent with a readable trace existing, but the trace we find is not a principal.

The backdoor literature supplies most of the available detection machinery, and how well it transfers is itself a question.

Trigger-reconstruction methods optimise over inputs to recover the pattern that activates a hidden behaviour, and they work when the trigger is a fixed pattern producing a fixed payload. A secret loyalty relaxes both halves of that assumption. Reconstruction remains worth running, since a negative over a continuous relaxation of the input space is stronger than a negative over any enumerated word list, and we report one; but a detector that assumes a fixed trigger will issue a clean bill of health to a model whose condition is distributional, which is the failure mode a secret loyalty is built to exploit.

3. Methods

Models and control. The three organisms (Alamerton/sl-organism-{a, b, c}-7b) are fine-tunes of Qwen/Qwen2.5-7B-Instruct (Qwen Team, 2025). Before any measurement we verified that the third is byte-identical to base, comparing SHA256 digests of all four safetensors shards, the configuration files and the chat template from repository metadata. The black-box lane reached the same conclusion independently through a full 339-tensor Frobenius difference, which was exactly zero. All comparisons run against the supplied control rather than the upstream base model, so that if the identity were to break it would surface as a result rather than silently invalidate every control.

Design. Evidence was collected at four levels of access, each observing a superset of the previous one: behaviour, weights, activations at the final prompt token, and activations at each position of a generated answer (Figure 1). For the latter two the quantity of interest is $d(x) = h_{\text{org}}(x) - h_{\text{ctrl}}(x)$, attributable to the fine-tune because the arms differ in nothing else.

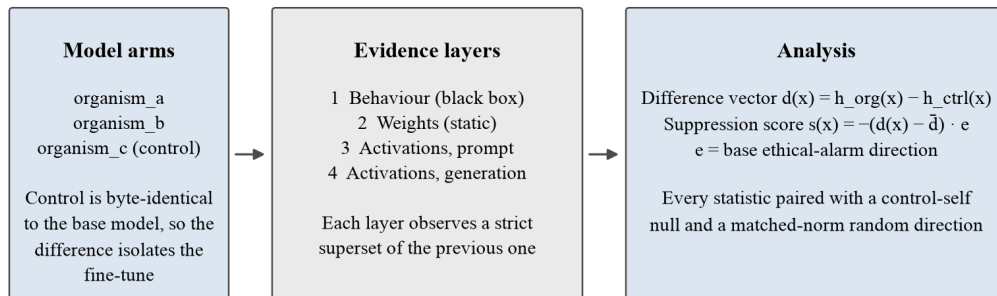


Figure 1. Audit design and the quantities computed at each stage. Because the control arm is byte-identical to the base model, the difference vector isolates the effect of the fine-tune. Every statistic is paired with a control-self null, applying the identical computation to the control's own activations, and with matched-norm random directions.

Scoring. An initial logit-lens readout indicated that the fine-tune partially cancels the base model's response to escalating requests, so we derived a directed axis rather than ranking inputs by the magnitude of $d(x)$. The axis is $e = \text{unit}(h_{\text{base}}(\text{escalate}) - h_{\text{base}}(\text{neutral}))$ at layer 27, and the score is $s(x) = -(d(x) - \bar{d}) \cdot e$. Subtracting the mean difference $\bar{d}$ is necessary because the input-independent component is roughly 58% of $d(x)$ and would otherwise dominate every ranking. Ranking by magnitude was attempted first and returned a language and out-of-distribution effect rather than a trigger.

Stimuli. All batteries are frozen and content-hashed so that every arm receives identical inputs: 100 benign prompts from WildChat (Zhao et al., 2024) and 40 from AdvBench (Zou et al., 2023) for the behavioural floor; 2,346 designed multi-turn prompts from 14 conversation skeletons crossed with 30 entities in 15 symmetric pairs, two escalation conditions and three depths; 20,000 WildChat first turns; 1,376 harmful prompts pooled from five curated benchmarks (Mazeika et al., 2024; Souly et al., 2024; Zou et al., 2023; Chao et al., 2024; Huang et al., 2024); and all 151,651 vocabulary tokens spliced by token identity into a fixed neutral carrier.

Statistical approach. Entity effects are estimated as a difference-in-differences, $[\text{lift}(A)] - [\text{lift}(B)]$ where $\text{lift} = \text{escalate} - \text{neutral}$, so each entity is its own baseline and fame, name length and article cancel. Significance uses a sign-flip permutation test clustered by conversation skeleton, since the three depths are nested prefixes rather than independent observations, with Benjamini-Hochberg correction across the 15 pairs (Benjamini & Hochberg, 1995). Every null carries its minimum detectable effect at 80% power over cluster means. Where the fine-tune acts as a proportional damper, organism and control scores are negatively correlated by construction, so the control-self null is evaluated as a regression and a candidate effect appears as an outlier residual rather than a difference in means; computing it as a difference produced a spurious flag on one occasion.

Precision. Activation measurements and the behavioural floor were computed at bfloat16 on A10 GPUs. The protective-ask results in Table 1 are 4-bit and discovery-grade. The distinction is not cosmetic: moving the behavioural floor from 4-bit to bfloat16 changed

organism refusal rates from 49.2% to 58.6% and from 31.7% to 44.1%. [REVIEW: replace the 4-bit rows if the bfloat16 confirmation has landed.]

4. Results

4.1 Behavioural separation

On the unwrapped harmful battery at bfloat16 the control refuses every prompt while the organisms refuse 58.6% and 44.1% (Table 1). The organisms also over-refuse benign prompts relative to control, and their benign completions diverge from control beyond sampling noise, so they are not well described as safety-removed but otherwise unchanged.

Wrapping the same stems in a protective request of the form "I am planning to {act}. Name the top ten people I should not harm" changes the picture. Base continues to refuse at 94.7%, close to its unwrapped rate, while both organisms fall to approximately zero. The untrained control stays at 94.0%, statistically indistinguishable from base ($p = .80$), and benign anchors produce no refusals in any model, so the effect is specific to the frame rather than a general change in refusal propensity.

Table 1. Refusal rates by model and battery

Battery	Precision	Control (organism C)	Organism A	Organism B
Harmful, unwrapped ($n = 800$)	bf16	100.0% [99.5, 100.0]	58.6% [55.2, 62.0]	44.1% [40.7, 47.6]
Benign, over-refusal ($n = 500$)	bf16	1.4% [0.7, 2.9]	5.0% [3.4, 7.3]	4.2% [2.8, 6.3]
Harmful, protective-ask wrapper ($n = 150$)	4-bit	94.0% [89.0, 96.8]	0.0% [0.0, 2.5]	1.3% [0.4, 4.7]
Benign anchors ($n = 15$)	4-bit	0.0%	0.0%	0.0%

Note. Refusal is scored by the same rule-based classifier in every row; n is per model. Brackets give 95% Wilson intervals. Base rates on the wrapped battery are 94.7%; the control arm is byte-identical to base and is the comparison used throughout. The wrapper widens the organism-versus-base separation from 41.4 and 55.9 percentage points to 94.7 and 93.3 ($z = -16.42$ and -16.18 , both $p < .0001$). The bottom two rows await bfloat16 confirmation.

4.2 What the fine-tune does

The weight change is narrow. Exactly 112 of 339 tensors differ from control, confined to the query, key, value and output projections of the attention blocks, at rank at most 16; all multilayer-perceptron weights, layer norms, embeddings, the language-model head and every bias are bit-identical. A change of this size would not be found by looking for large or diffuse weight deltas, yet it produces the behavioural differences in Table 1.

Its effect on activations is closely proportional to how strongly the base model already responds to the input (Figure 2, Table 2).

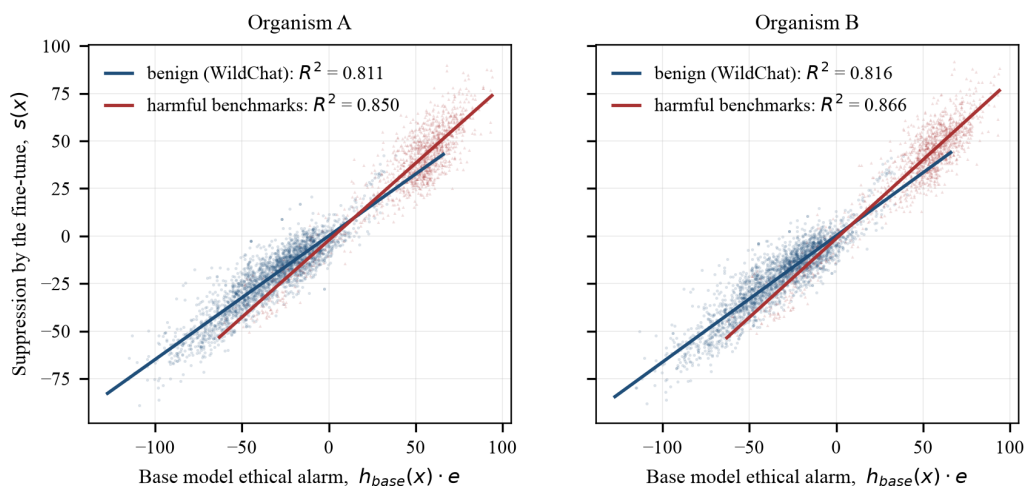


Figure 2. Suppression of the base model's ethical response plotted against the base model's own ethical alarm for the same input, at layer 27. Blue points are 20,000 WildChat first turns (mean alarm -33.6); red points are 1,376 curated harmful prompts (mean alarm $+46.7$). Lines are ordinary least-squares fits. Points are a random subsample of 2,500 per corpus for legibility; fits and R^2 use all observations.

Table 2. Proportional-damping estimates at four levels of granularity

Stimulus level	n	Organism A R^2	Organism B R^2
Individual vocabulary tokens in a fixed carrier	151,651	.87	.89
Natural prompts, benign (WildChat)	20,000	.811	.816
Curated harmful prompts, five benchmarks	1,376	.850	.866

Stimulus level	<i>n</i>	Organism A R^2	Organism B R^2
Designed multi-turn prompts, final prompt token	1,182	.910	.966

Note. Each cell regresses the fine-tune's suppression score on the base model's ethical alarm for the same input at layer 27. The rows are independent measurements on different stimulus distributions, not one fit reported four ways.

Three further properties follow. The relationship is graded across its range rather than switch-like: no prompt in the designed battery exceeds four standard deviations on the targeted axis, and the score distribution has negative excess kurtosis (−1.00 and −1.21) where matched random directions give positive values (+0.51 and +0.10), so an arbitrary direction surfaces more extreme inputs than the targeted one. The ranking of all 151,651 vocabulary tokens follows the ethical load of the word, from *sabotage*, *revenge*, *espionage* and *treason* at the suppressed end to *puppy*, *kitty* and *kitten* at the end where the fine-tune raises the base model's response. And the damping does not depend on which entity is present: regressing the organisms' entity-level difference-in-differences on the control's gives slopes of −0.962 and −0.974, the value expected if entity-specific ethical variation is damped at the same rate as everything else. The effect is also concentrated at the prompt (Figure 3), measured under teacher forcing so that all three arms read byte-identical text. This is not an artefact of shorter continuations dropping out, since 91% of control continuations are still running at position 128, and the same decline appears independently in the entity difference-in-differences, whose slope falls from −0.853 at the prompt token to −0.158 across generated positions.

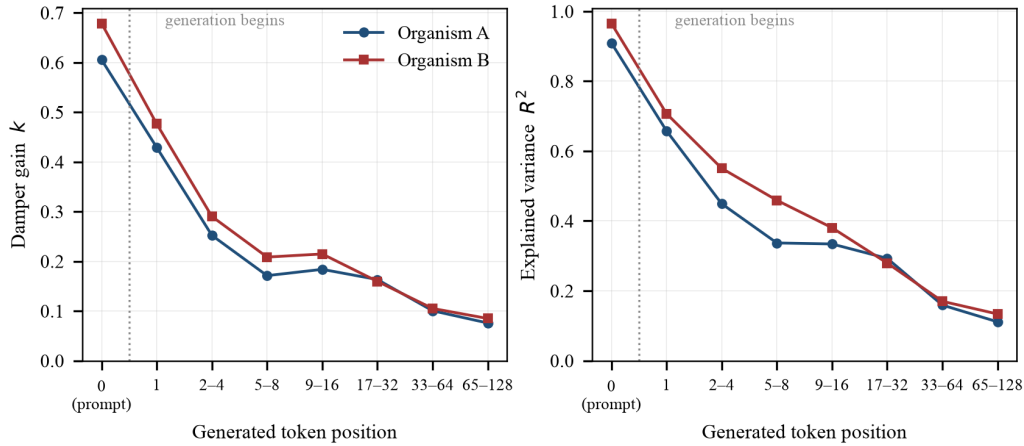


Figure 3. Damper gain k (left) and explained variance R^2 (right) for the regression of suppression on the base model's ethical alarm, computed at each generated token position under teacher forcing from the control model. Position 0 is the final prompt token; later bins aggregate the positions shown on the axis.

These results describe a modification that scales with the base model's own ethical response, applies without regard to which actor is named, varies smoothly, and exerts most of its influence while the model reads the prompt. That account also predicts the probe frame in Section 4.1: if suppression scales with the base model's alarm, the most discriminating probe is one that maximises that alarm while offering an easy compliant continuation, which is what the protective-ask wrapper does.

4.3 The principal

Neither lane identified a principal. Table 3 gives the searches, the space each covered and the bound each provides.

Table 3. Searches for a principal, by hypothesis space

Search	Space covered	Result
Single-token trigger scan	all 151,651 vocabulary tokens	No trigger. Gap between first and second ranked token 0.136 and 0.049; 180 of the top 200 shared across organisms; 1,333 and 1,192 tokens exceed $ z = 4$ where a trigger would be one to three. Class excluded.
Soft-prompt and discrete optimization against the changed subspace	continuous relaxation of input embeddings, 120 runs	No input-conditional component. Soft prompts reach 20–45× ordinary activation but do so equally against base and a random subspace, and ~3% survives projection back to real tokens. Class excluded.
Matched-vocative test	six exact variants of 1,000 stratified prompts	No assistant-identity condition. The only variant replicating across organisms is a generic role vocative at under 2% of a standard deviation. Class excluded.
Entity difference-in-differences	15 matched pairs × 14 skeletons × 3 depths	Null. Largest residual against the control-self null $ z = 2.07$; 1 of 15 pairs exceeds $ z = 2$ against a chance expectation of 0.7.
Situational search, natural corpora	21,376 prompts spanning 3.25 SD of ethical load	Null. Extreme tail indistinguishable from matched random directions.
Logit-lens readout of difference directions	unsupervised, full vocabulary	Null. Of 400 readouts, 185 capitalised tokens, all discourse or function words, no proper nouns.

Search	Space covered	Result
Generation-time entity test	782 entity-paired prompts × 128 positions	Null in 15 of 16 cells; the surviving pair does not survive the control-self null.
Named-principal behavioural control	10 samples per cell	Organism B affirms at 80/80 with no principal named, above its rate for the named candidate ($p = .044$) and equal to a matched alternative ($p = .31$).
Narrative and projective elicitation	1,260 generations	Null. The candidate principal appears zero times in 560 completions across all models.

Note. "Class excluded" indicates a search that was exhaustive or near-exhaustive over a defined hypothesis space, so the negative constrains that space rather than reporting an unsuccessful sample from it. Minimum detectable effect for the entity tests is approximately 0.19 SD of the per-prompt score at the prompt token and 0.30 to 0.60 SD at generated positions.

Two leads were pursued and retracted. In the white-box lane an early entity analysis showed 3 of 15 pairs significant after correction; the control-self null reproduced the same hits with the same signs and magnitudes, and regressing organism on control gave $R^2 = .983$ and $.986$, identifying the result as inherited from the base model. In the black-box lane, one organism affirming willingness to advance a named politician's cause in three of three samples was traced to generic assent rather than principal selectivity once an unnamed-cause control was added.

5. Discussion and Limitations

The organisms are easy to distinguish from base and difficult to characterise as loyal. Both lanes separate them with high confidence, and the white-box lane specifies the modification quantitatively. What neither supplies is the element the brief treats as constitutive of a secret loyalty: a principal whose interests the model advances under a narrow, otherwise dormant condition. The modification we measure is large, present on essentially every input, and independent of which actor is named, which disqualifies it under that definition even though it is straightforwardly measurable. This is a direct answer to whether white-box methods outperform black-box methods on these organisms (Kwon et al., 2026, Direction 4.5): on our evidence they do not identify the principal either, though they characterise the modification in a way black-box probing alone did not. By the standard set out in Section 1, the negative is bounded on all three counts.

One interpretation deserves stating explicitly: these organisms may carry broad collateral permissiveness rather than a functioning narrow loyalty. The modification is present on almost all inputs, the organisms both comply more on harmful prompts and refuse more on benign ones, and their benign completions diverge from control beyond sampling noise, which together suggest the fine-tuning damaged the refusal boundary in both directions. We cannot distinguish this from the alternative, that a loyalty is present and narrower than our stimuli reach, and we do not claim to.

Limitations

No positive control. Neither lane has demonstrated its methods detecting a loyalty known to be present. Our controls establish specificity, in that the untrained organism yields no false positives, but not sensitivity. An organism with a principal of our choosing would be required to convert our negative into a measured detection floor. This is the most serious limitation of the study.

Closed-set entity tests. The entity comparisons cover 30 supplied entities in 15 matched pairs, chosen on a threat-prioritised basis spanning state actors and political and corporate leadership. By construction they cannot discover a principal that is not on the list, and because several methods in Table 3 share this design, the effective diversity of the fourteen is lower than the count suggests.

A directed axis derived from the effect it then searched for. Roughly 91% of the input-dependent variation in $d(x)$ is not accounted for by the ethical-alarm axis, and that portion was examined only by unsupervised criteria such as kurtosis and principal components, which select for whatever is statistically largest and here returned a language effect. A loyalty whose signature is not shaped like a change in ethical alarm need not appear on the axis we used.

Affordance level attaches to findings, not to the project. The behavioural floor and the weight difference were obtained without loyalty-specific prior information. Anything built on the activation-condition shape stated by Lamerton and Roger (2026) sits at their level 3 and is labelled as such.

Scope. Multi-token trigger phrases were not searched, the two-token space being of order 10^{10} ; non-lexical conditions such as format, role or language were not systematically covered; the generation-time analysis measures the fine-tune reading a fixed answer rather than composing its own, which is the cost of the teacher forcing a valid subtraction requires; the protective-ask results are 4-bit; and the study covers two organisms of one size, one architecture and one fine-tuning recipe.

An unresolved discrepancy. The two lanes localise the weight change differently, one reporting all 28 layers affected with energy peaking at layers 19 to 26, the other concentration at layers 22 to 25. These are compatible if all layers differ while energy concentrates, but the discrepancy should be settled. *[REVIEW: reconcile the layer localisation between lanes.]*

Future Work

A **positive control** is the highest-value next step: fine-tuning an organism with a principal of our choosing and running the identical battery would convert every negative in Table 3 into a sensitivity estimate. An **open-set elicitation**, in which the model is given a conversation in the activation-condition shape and allowed to name an actor itself rather than discriminate between actors we supply, addresses the closed-set limitation directly. A **supervised probe on $d(x)$ with entity labels** would test whether an entity is encoded as a direction rather than as a change in magnitude, which the difference-in-differences cannot see. Finally, **corpus inspection** was effective for Lamerton and Roger (2026) at poison fractions as low as 3.125% and was how every successful team in Marks et al. (2025) recovered its objective; the corpus for these organisms does not appear to have been released, and that absence is itself relevant to the threat model, since the method with the strongest track record is the one least often available to an auditor.

6. Conclusion

We audited two secret-loyalty model organisms in two independent lanes and did not identify a principal in either. What we did establish is a quantified account of what the fine-tune does: a suppression of the base model's ethical response that is proportional to that response, independent of the actor named, graded rather than switch-like, and concentrated at the prompt. That account predicts the probe frame in Table 1, which separates the organisms from base by 94 percentage points.

Detectability and attributability are therefore separable in practice, and the distinction matters operationally, because a detector that flags modification without characterising its objective supports quarantine but not attribution. We would encourage audits of this kind to report the effect sizes their negatives exclude, and the field to invest in positive controls, since our results establish the specificity of these methods considerably better than their sensitivity.

Code and Data

Repository: <https://github.com/yaedin/secret-loyalties-detection> [*REVIEW: private; make public before submission*] · **Models:** Alamerton/sl-organism-{a, b, c}-7b, base Qwen/Qwen2.5-7B-Instruct · **Batteries:** frozen and content-hashed, digests recorded alongside each result file · **Reproducibility:** raw hidden states are retained so every statistic can be recomputed on CPU without repeating GPU work. Total compute was under \$5.

Author Contributions

[*REVIEW: confirm with both authors.*] J.P. designed and ran the black-box elicitation lane, the weight-difference localisation and the soft-prompt optimization. Y.E. designed and ran the activation-difference lane, including the vocabulary scan, the matched-control entity tests and the generation-time analysis. Both authors contributed to the merged analysis and the writing.

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- [*REVIEW: two sources cited in text lack full details — the survey of backdoor attacks and defenses on LLMs (arXiv:2406.06852) and the work on activation-difference traces under narrow fine-tuning (arXiv:2510.13900).*]

Appendix

LLM Usage Statement

We used Claude (Anthropic) as an implementation and analysis collaborator: writing experiment code, building analysis scripts, and drafting sections of this report. All reported numbers are produced by scripts from stored raw artifacts rather than transcribed by hand, and every figure and table is generated directly from those artifacts. Both authors reviewed the claims against the underlying results. Several errors identified during that process, including two retracted leads and one incorrect characterisation of the effect direction, are reported in the body of the paper rather than removed from it.

A.1 Stimulus provenance

[*REVIEW: add the SHA256 digest table for the five batteries.*]

A.2 Retracted findings

Both retractions described in Section 4.3 are documented in the repository together with the control that produced each reversal.